Vector spaces

A set V with field F and operations $+$ and $\cdot$ such that $\forall u, v, w \in V, \forall a, b \in F$:

- $u + v \in V$
- $a \cdot u \in V$
- $u + v = v + u$
- $(u + v) + w = u + (v + w)$
- $\exists 0 \in V : u + 0 = u$
- $\exists (-u) \in V : u + (-u) = 0$
- $1 \cdot u = u$
- $(ab) \cdot u = a \cdot (b \cdot u)$
- $(a + b) \cdot u = a \cdot u + b \cdot u$
- $a \cdot (u + v) = a \cdot u + a \cdot v$

subspace a subset S of V that is itself a vector space under the same operations. It is sufficient that $S \neq \varnothing$ and $\forall u, v \in S \, \forall a, b \in F, au + bv \in S$.

linear combination for a subset S of V, a sum of scalar multiples of vectors in S, $\sum_{v \in S} a_v v$

linear independence the property of a subset S of V where $\sum_{v \in S} a_v v = 0 \implies \forall v \in S, a_v = 0$

span $\operatorname{Span}(S) := \left\{ \sum_{v \in S} a_v v \,\middle|\, \forall s \in S, a_s \in F \right\}$, the set of all linear combinations of vectors in S.

basis a subset B of V that is linearly independent and spans V.

dimension $\dim(V) = |B|$ for basis B.

sum of vector subspaces for subspaces U and W of vector space V, $U + W := \{u + w \,|\, u \in U \wedge w \in W\}$

Theorems

- Zero vectors are unique for each vector space.
- Each vector has a unique additive inverse.
- Intersections of subspaces are subspaces.
- Spans are subspaces.
- Every vector space has a basis.
- Every vector has a unique representation w.r.t. a basis.
- Every basis of a vector space has the same cardinality.
- for a subset S of V, $|S| > \dim(V) \implies S$ is linearly dependent.
- for a subset S of V, the following each imply S is a basis:
  - S is linearly independent and spans V
  - S is linearly independent with $|S| = \dim(V)$
  - S spans V with $|S| = \dim(V)$
- for subspaces W_1 and W_2, $\dim(W_1 + W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2)$

Standard bases

F^n $\left\{ (\delta_{ij})_{1 \leq i \leq n} \,\middle|\, 1 \leq j \leq n \right\}$, the set of tuples where one term is 1_F and the rest are 0_F.

$M_{m \times n}(F)$ $\{E_{ij} \,|\, 1 \leq i \leq m \wedge 1 \leq j \leq n\}$, the set of matrices where one entry is 1_F and the rest are 0_F.

$P_n(F)$ $\left\{ x \mapsto x^i \,\middle|\, 1 \leq i \leq n \right\}$

square symmetric $n \times n$ matrices $\{E_{ii} \,|\, 1 \leq i \leq n\} \cup \{E_{ij} + E_{ji} \,|\, 1 \leq i < j \leq n\}$

Linear maps

For vector spaces V and W over field F, a map $f : V \to W$ such that $\forall a, b \in F \, \forall u, v \in V = f(au + bv) = af(u) + bf(v)$.

kernel $\operatorname{Ker}(f) := \{v \,|\, v \in V \wedge f(v) = 0\}$ is the set of vectors in V that f takes to the zero vector in W. Also called the null space or notated $\mathrm{N}(f)$.

nullity $\operatorname{null}(f) := \dim(\operatorname{Ker}(f))$

image $\operatorname{Im}(f) := \{w \,|\, w \in W \wedge \exists v \in V : w = f(v)\}$ is the set of outputs of f in W. Also called the range or notated $\mathrm{R}(f)$.

rank $\operatorname{rank}(f) := \dim(\operatorname{Im}(f))$

matrix of a linear map If $B := \{b_i \,|\, 1 \leq i \leq \dim(V)\}$ is an ordered basis of V and $C := \{c_i \,|\, 1 \leq i \leq \dim(W)\}$ an ordered basis of W, then there exists the matrix $[f]_B^C = \left[f(b_1) \cdots f\left(b_{\dim(V)}\right) \right]$ such that $f(v) = [f]_B^C v$.

coordinate vector $[v]_B := \left[v_1 \cdots v_{\dim(V)} \right]^\intercal$, given that $v = \sum_{i=1}^{\dim(V)} v_i b_i$

injective if $f(x) = f(y) \implies x = y$. Synonymous with one-to-one.

surjective if $\forall y \in W, \exists x \in V : y = f(x)$. Synonymous with onto.

isomorphism f is surjective and injective. Equivalently invertible.

Theorems

- $f(0_V) = 0_W$
- f is injective $\iff \operatorname{Ker}(f) = \{0\} \iff \operatorname{null}(f) = 0$
- f is surjective $\iff \operatorname{Im}(f) = W \iff \operatorname{rank}(f) = \dim(f)$
- The inverse of an isomorphism is an isomorphism.
- Vector spaces are isomorphic iff they have the same dimension.
- All linear maps preserve linear dependence.
- Injective linear maps preserve linear independence.
- If V is finite dimensional, then $\operatorname{null}(f) + \operatorname{rank}(f) = \dim(V)$
- The basis of $\operatorname{Ker}(f)$ can be found by finding the RREF A of $[f]$ and solving $Ax = 0$ for the pivot variables in terms of the free variables, substituting these back into x and factoring out the free variables. The basis is then the set of vectors in this linear combination.

- The basis of $\mathrm{Im}(f)$ is the set of columns of $[f]$ that are pivot columns in RREF.
- Rank of a linear map f can be found by putting $[f]$ in RREF and counting the pivots.
- Nullity of a linear map f can be found via the previous theorem and the rank-nullity theorem.

Matrices

addition same dimensions, entrywise

scalar multiplication entrywise

matrix multiplication if A is an $m \times p$ matrix and B is an $p \times n$ matrix, then the product AB is an $m \times n$ matrix defined by $[AB]_{ij} = \sum_{k=1}^{p} a_{ik} b_{kj}$. In other words, $[AB]_{ij}$ is the dot product of the ith row of A and the jth column of B.

inverse for square matrix A, the square matrix A^{-1} such that $AA^{-1} = A^{-1}A = I$.

transpose $[A^\intercal]_{ij} := [A]_{ji}$

Theorems

- A matrix is invertible iff the only solution to $Av = 0$ is $v = 0$ iff the determinant is nonzero.
- $Ax = b$ has a solution (is consistent) iff $\mathrm{rank}(A) = \mathrm{rank}(A|b)$
- if $Ax = b$ is consistent with rank r, every solution is $s = \sum$